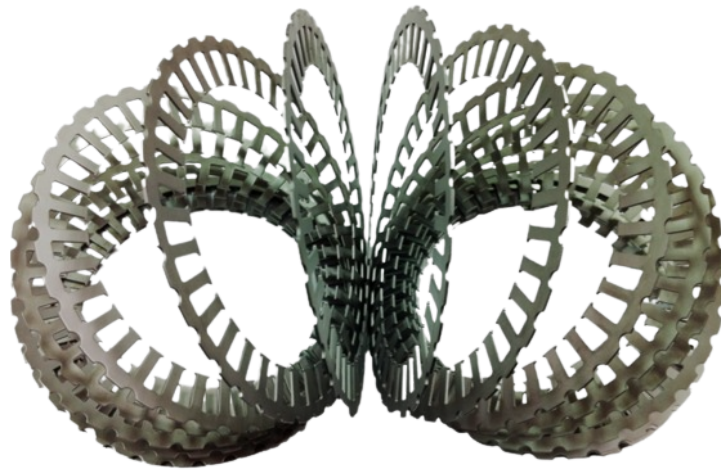


Feasibility of the Manufacture of Slinky Stator Geometries

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Introduction

Background

The slinky method of stator manufacture consists of coiling a cut strip of metal into a stator instead of stacking laminate sheets. This saves material which would otherwise be wasted in the centre cutout of the circular stator shape.

The objective of this study is to conduct non-linear FEA analyses on certain stator geometries to determine the feasibility of manufacturing using the slinky method.

Scope

The following studies will be conducted:

- Study 1: Analytical solution to find the minimum OD for a stator
- Study 2: Non-linear FEA of back iron under a bending moment
- Study 3: Non-linear FEA of varied stator geometries under bending moment

Geometry

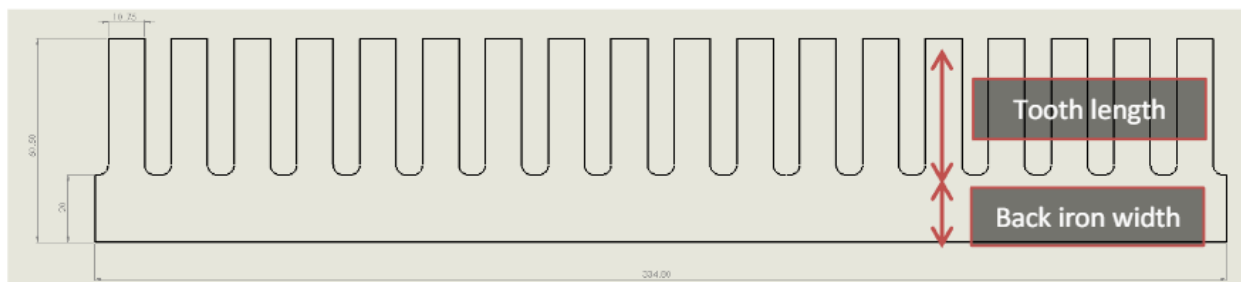


Figure 1 - Slinky stator geometry [1]

The simplest geometry comprises a rectangular back iron and rectangular teeth, as illustrated in Figure 1. This means that the primary dimensions are the back iron width, and the tooth width, length and spacing. Other features like fillets, chamfers, and notches will be explored in Study 3 and Study 4.

In these studies, simulating under 5 teeth is sufficient due to the repetitive strip pattern and uniform bending induced by the forming process.

Materials

Properties

The material used for this study is **NO20 Electrical Steel** with the following properties:

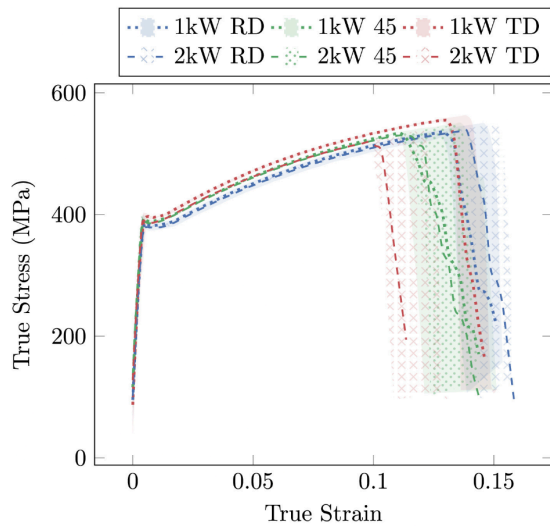


Figure 3 - Stress-strain curve [2]

Density:	7.65 kg/dm ³
Young's Modulus:	185 GPa
Poisson's ratio:	0.33
Yield Stress:	400 MPa
Strain at fracture:	0.12-0.17
UTS:	530 MPa
Values for the yield strength, tensile strength and Young's modulus are given for the rolling direction. Corresponding values for the transverse direction are approximately 2 % higher.	

Assumptions

The following assumptions will be made for the FEA simulations

- For the plastic region of deformation, a bilinear model will be used to save computation time as the stress-strain graph is linear enough to assume sufficient results.
- Given the material is non-oriented, it shall be assumed to be isotropic.
- Thermal properties will also be neglected and the FEA will be at room temperature.
- Out-of-plane displacement shall be neglected (including large buckling behaviours) as the manufacturing method should provide sufficient support to prevent movement during the coiling process.

Ansys Bilinear Model

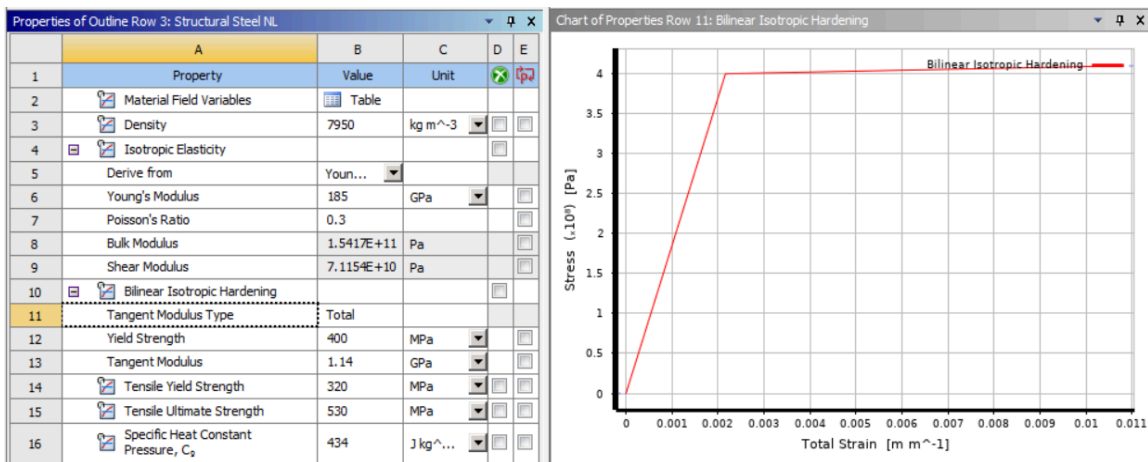


Figure 4 - Bilinear material setup

Study 1: Analytical Solution

Objective

This analysis will provide a baseline performance for the rest of the studies. A stator without any teeth will demonstrate uniform behaviour under bending moments, therefore it can be used to find a relationship between the minimum OD of a stator and its back iron width.

Solution

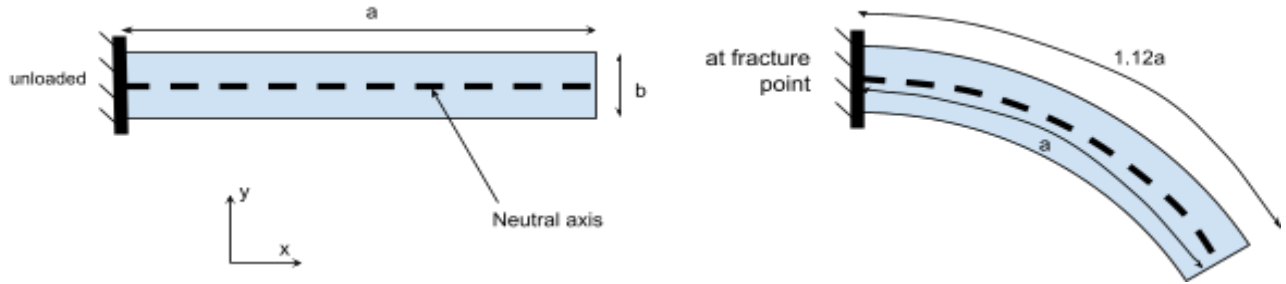


Figure 5 - Problem setup

Given the strip is in the plastic deformation region, the analysis becomes more complicated if the standard cantilever beam analysis is used. Therefore instead the 'Strain at fracture' property can be used to determine the point of failure. If there is no buckling due to the manufacturing process constraining the strip to a plane, it is logical to assume the failure will be a fracture along the outermost edge, as the most elongation occurs there.

The strip will fail where the outer edge is 12% longer than the neutral axis. Using the arc length formula the radius of curvature can be calculated, and as they are assumed equal, a relationship between the width and the radius of curvature can be found.

$$\begin{aligned} R_{min} \cdot \theta &= a \\ (R_{min} + b/2) \cdot \theta &= 1.12a \\ R_{min} &= 4.167b \\ \Rightarrow OD_{min} &= 9.33b \end{aligned}$$

Figure 5 can be used as a 'best case' to compare the results of later studies against.

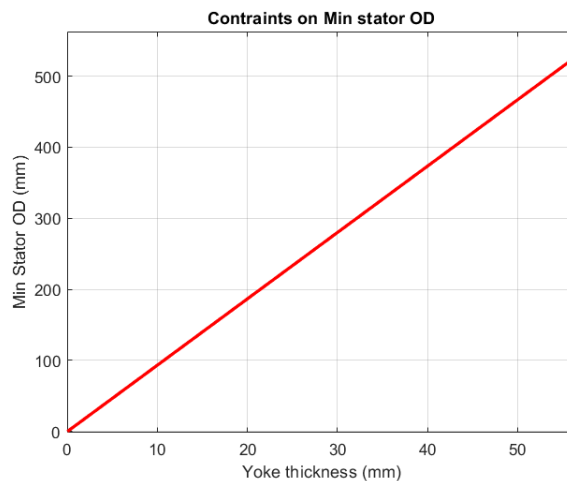


Figure 6 - Analytical baseline graph

Study 2 - FEA on Back Iron

Objective

This study will test how representative the analytical solution is of the strip's behaviour. A number simulations with varying back iron widths will be run and plotted on the baseline graph. It is expected to see a close correlation for small thicknesses, but as the strip gets thicker, it will likely fail at larger radii.

Model

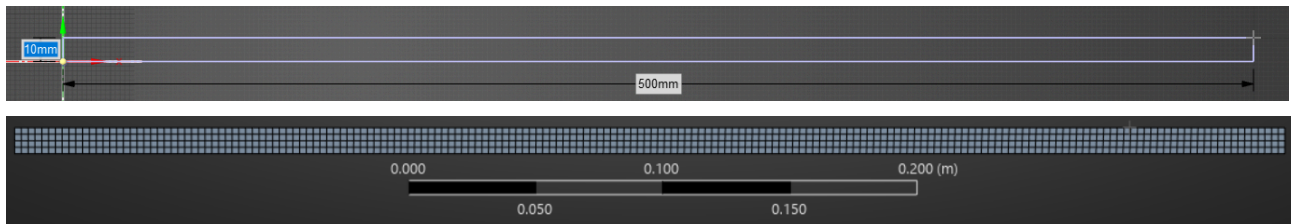


Figure 7.1 - Geometry and mesh

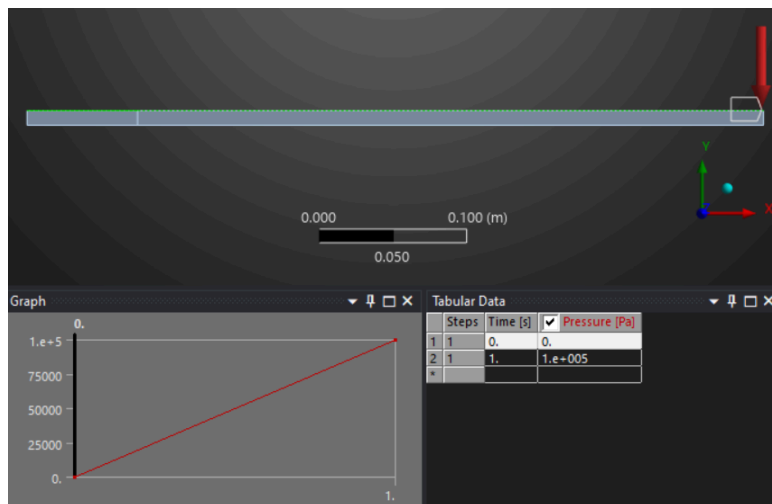


Figure 7.2 - Load applied to the beam

The boundary conditions consist of a pressure on the outer surface to simulate the bending force from the coil former, and a fixed edge on the left.

The pressure increases over time so the time that the material reaches its strain limit can be found. The maximum pressure is selected to ensure failure is reached.

Results

The displacements of the outer edge of the strip are taken when either the material is at the safe UTS or the safe fracture strain limit, then the radius of curvature can be found by smoothing positions with a moving average, then using the 2D curvature formula [3].

The FEA setup is not equipped to model failures in the form of rippling, microcracking, or any other types of compression-induced failures. Choosing a strain safety factor of 1.2 to neglect the conditions where these effects start to occur ensures the model remains as accurate as possible.

After finding the radius, the minimum OD can then be plotted on the baseline graph to validate the results of this study. As seen in Figure 9 the values align closely at small thicknesses, but diverge at larger thicknesses. This validates the Study 1, demonstrating manufacture of stators in the region below the baseline is infeasible.

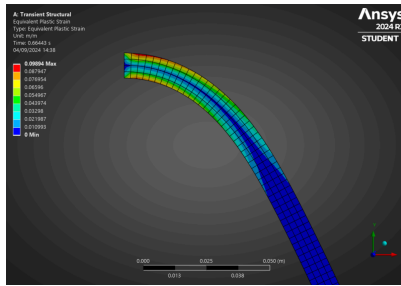


Figure 8.1 - 10mm strip

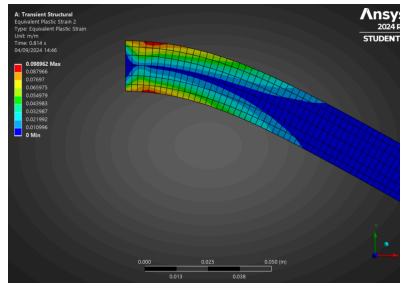


Figure 8.2 - 20mm strip

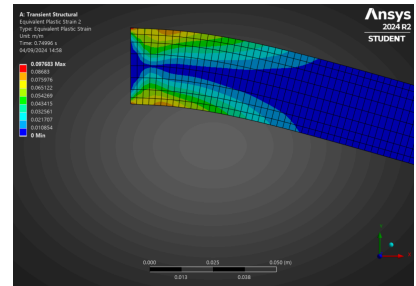


Figure 8.3 - 30mm strip

Width (mm)	Min Radius (mm)	Min OD (mm)	Width / Min OD
5	27.50947	55.01894	0.090878
10	56.89503	113.7901	0.087881
15	82.87266	165.7453	0.0905
20	112.2967	224.5935	0.08905
25	147.7196	295.4392	0.08462
30	170.9251	341.8501	0.087758
35	202.9266	405.8532	0.086238
40	257.2953	514.5907	0.077732
45	326.2923	652.5845	0.068957
50	430.6304	861.2607	0.058054
55	657.1989	1314.398	0.041844
60	1020.266	2040.531	0.029404

Table 1 - Study 2 results

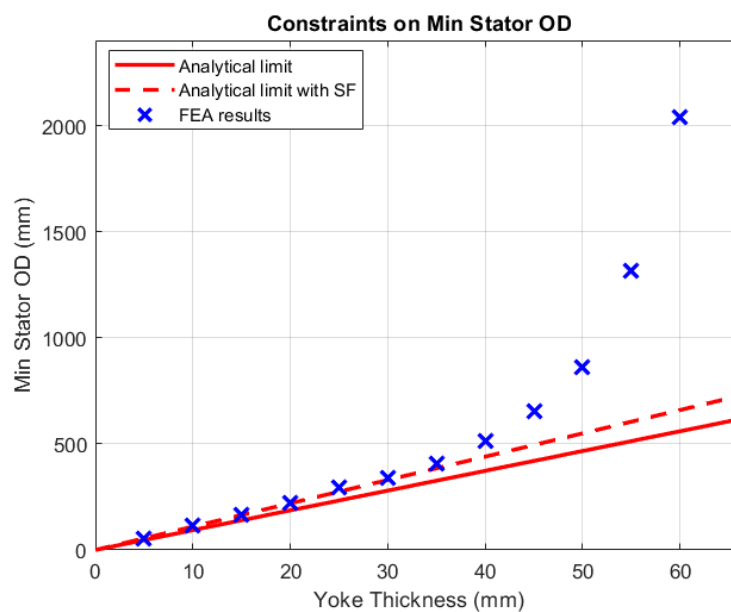


Figure 9 - FEA results compared to analytical baseline

Study 3 - The Effects of Teeth

Objective

The purpose of this study is to validate the hypothesis that a slinky stator without teeth can always coil more tightly than a stator with teeth. This assumption is based on the idea that, without teeth, the stress distribution is more uniform across the back iron, whereas the presence of teeth causes stress to concentrate in specific areas, leading to earlier failure.

Model

Due to the many configurations of stator geometry, tests will only be run on a select few representative stators, with the parameters highlighted in Figure 10 being adjusted. The tooth length shall be long enough to neglect the effects on the failure point.

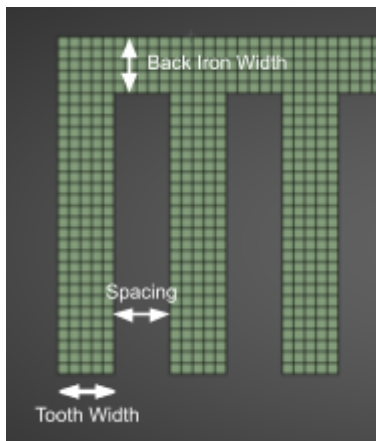


Figure 10 - Parameterised geometry

Back Iron Width (mm)	Spacing (mm)	Tooth Width (mm)
10	6	6
	8	8
	10	10
20	12	12
	16	16
	20	20
30	18	18
	24	24
	30	30

The radius shall be captured in a different way to Study 2 as the teeth will introduce a non-uniformity to the curvature. It can be found by taking position samples around the point of failure and then finding the average radius, as opposed to the minimum.

Results

The results of this study successfully validated the hypothesis that a slinky stator without teeth can coil more tightly than a stator with teeth. As expected, The presence of teeth caused stress to concentrate in localised areas, leading to earlier failure and a reduced coiling capacity.

It was also found that larger teeth lead to a larger radius, this was due to the increase in support in the back iron adjacent to the teeth, causing the curvature to tend more to a polygonal shape instead of a circle. Figure 11 demonstrates the localised strain

regions along the outer edge separating and reducing uniformity. Note that these images depict the model after it has failed, and are only for visualising the localised regions.

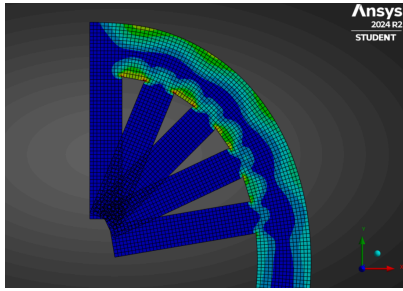


Figure 11.1 - 18mm teeth

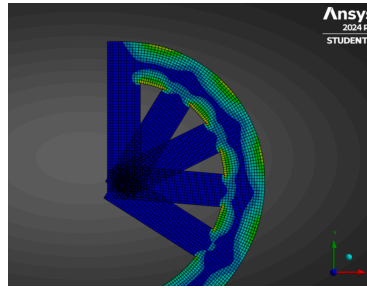


Figure 11.2 - 24mm teeth

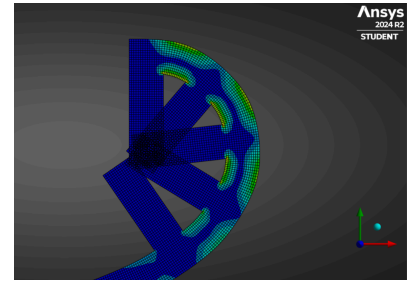


Figure 11.3 - 30mm teeth

Back Iron Width (mm)	Spacing (mm)	Tooth Width (mm)	Min Radius (mm)	Min OD (mm)	Width / Min OD
10	6	6	84.80641	169.6128	0.058958
	8	8	97.09062	194.1812	0.051498
	10	10	130.9429	261.8858	0.038185
20	12	12	162.9119	325.8238	0.061383
	16	16	221.9011	443.8022	0.045065
	20	20	302.7641	605.5283	0.033029
30	18	18	279.9324	559.8647	0.053584
	24	24	328.8068	657.6136	0.045619
	30	30	531.3108	1062.622	0.028232

Table 2 - Study 3 results

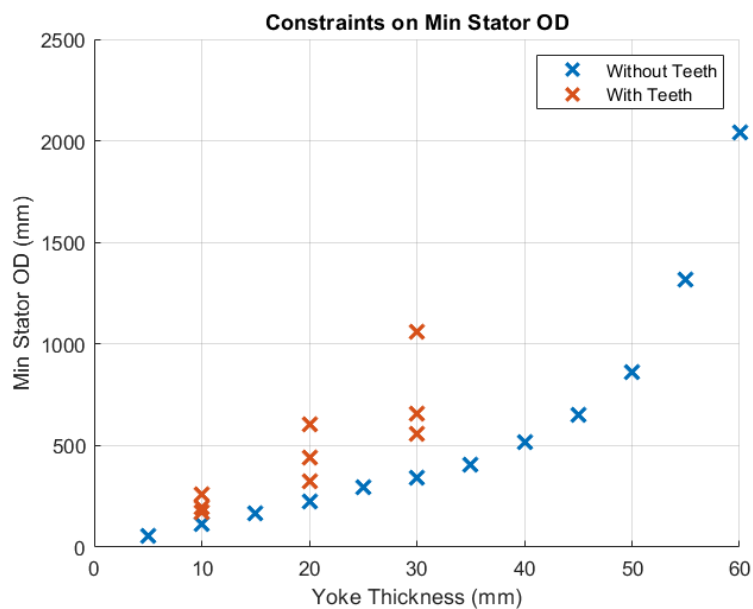


Figure 12 - Plot showing the effects of teeth

References

- [1] N. Dodd, "Feasibility of alternative manufacturing processes for winding and stator components of electric machines", no. 108-120, 2021.
- [2] L.M. Jones, A. Winter, L. Tinkler, G.W. Jewell, H. Ghadbeigi, "Influence of Remote Laser Cutting on Magnetic Loss and Mechanical Properties in 0.2 mm Silicon Steel", Journal of Magnetism and Magnetic Materials, Volume 607, 2024, ISSN 0304-8853,
- [3] Wikipedia contributors, 'Curvature', Wikipedia, The Free Encyclopedia, 28 August 2024